United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Mid Exam: Summer 2025

Course Code: CSE 2215, Course Title: Data Structures and Algorithms 1

Total Marks: 30 Duration: 1 hour 30 minutes

Answer all questions. Marks are indicated in the right side of each question.

[Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.]

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1. a. Suppose you want to sort each of the following three arrays in **ascending order using Quicksort**. You are considering the **last element** as pivot. Identify the array for which Quick Sort will face the worst case scenario. Show the simulation for **first partition of that array** and justify your answer. [3]

   i. 12, 34, 11, 98, 43, 63
   ii. 95, 87, 57, 11, 5, 38
   iii. 83, 75, 49, 36, 31, 25

   b. "Merge Sort requires more space than Quick Sort"-Is this statement True or False? Explain briefly. [2]

   c. Is it a good idea to use Counting Sort to sort an array of n integers in the range of [1, n^3]? Explain your answer. [2]

2. a. You have a 0-indexed array grid[50][100]. Every element of this array is a 4 byte integer. The element grid[17][25] is located at address 16900 and the element grid[32][1] is located at address 22804.

   Does this array follow row-major order memory addressing or column-major order memory addressing? Determine with proper justification. [4]

   Note: It is guaranteed that the addresses are correct and the memory addressing order can be determined conclusively.

   b. For each of the following arrays, determine the more efficient algorithm between Linear Search and Binary Search to find a given target value. [2]

   i. [2, 5, 8, 12, 16, 23, 38, 45, 56, 72]
   ii. [”apple”, ”banana”, ”blueberry”, ”cherry”, ”grape”, ”guava”, ”mango”, ”pear”]
   iii. [42, 7, 19, 85, 13, 27, 66, 34]
   iv. [99, 87, 74, 63, 51, 40, 29, 15]

3. a. Find the exact time equation for the following algorithm and represent it using Asymptotic Tight Bound Notation. [3]

```
for(int i = 1; i<=n; i=i*2) {
        for(int j = 1; j*j<=n; j++){
                s = s + i*j;
        }
}
```

b. Assume two algorithms to complete a specific task take $O(n^2)$ and $O(n\log n)$ times, respectively. However, the first approach takes $O(n\log n)$ space while the second approach takes $O(n^2)$ space. Which algorithm should you use and why? Justify your answer. [2]

4. a. Explain a situation where an **array** is more suitable than a **linked list**, and justify your answer with one clear reason. [2]

b. Consider a task scheduling system that maintains a list of processes to be executed sequentially. New processes may arrive at any time and must be added to the end of the list, while execution always begins with the first process, and completed processes are removed from the system. Explain the advantage of maintaining both **head** and **tail** pointers in a linked list for efficiently managing such a system. [2]

c. You are given the **head of a sorted singly linked list** and a value **key** that needs to be inserted into the list. Write pseudocode for a function that inserts the key into its correct position so that the list remains sorted after insertion. [3]

Example:
If the linked list is 10 → 12 → 18 → 25 and the key is 15, then after insertion, the list should become: 10 → 12 → 15 → 18 → 25

5. a. Suppose you have implemented a queue using a circular array of size 3, and it already has two elements enqueued, so the array looks like this:

12	15	

front = 0, rear = 1

Execute the following operations on this queue, and after every operation show the current status of the array, and the values of front and rear. [3]

i. dequeue()
ii. enqueue(8)
iii. enqueue(17)
iv. dequeue()
v. dequeue()
vi. dequeue()

b. Suppose you have implemented a stack using a singly linked list, and you are maintaining both the head and last pointers of the list. What should be the point of push and pop in this implementation for optimal time complexity – head or last? Explain briefly. [2]